TubigLoop CDO: Closed-Loop Drainage Obstruction Triage, Verified Solid Waste Diversion, and Rainwater Reserves

A Comprehensive System Concept, Operational Architecture, and Policy Briefing for Concept Paper Authors, Grant Proposals, and Institutional Stakeholders.

Target Program: CEIW 2026 Green Innovation / DOST-PCIEERD	**Target Geography:** Cagayan de Oro City (Misamis Oriental, Region X)
Primary Domain: Urban Drainage DRRM & Circular Economy	**Lead Deployment Pilot:** USTP CDO Campus & Bitan-ag Creek Corridor
Technology Stack: Next.js 16, Supabase RLS, Gemini Vision AI, Leaflet GIS	**Document Version:** 1.0 (Production Architecture Specification)

1. Executive Summary & Core Value Proposition

TubigLoop CDO is an integrated, mobile-first digital orchestration and physical chain-of-custody platform engineered to mitigate the chronic urban and campus monsoon flash-flooding crises across Cagayan de Oro City. The platform coordinates photographic obstruction reporting by citizens and students, executes automated multimodal Vision AI waste categorization, performs transparent risk-based triage dispatching, enforces verified on-site scale weighing by material fraction, and closes the loop through formal handovers to accredited Materials Recovery Facilities (MRFs) and recyclers.

> **Core Operational Axiom:**
> *"Software ra atong product. Atong trabaho kay klarohon ang report, assignment ug evidence. Ang actual cleanup ug recycling kay sa participating campus personnel ug partner recyclers."*
> (English translation: *Our software product provides absolute clarity in reporting, risk prioritization, task assignment, and physical chain-of-custody evidence. Physical desiltation, clearing, and recycling are executed by authorized campus personnel, LGU maintenance crews, and licensed recycling cooperatives.*)

Fundamental Value Pillars for Institutional Stakeholders:

- **End-to-End Chain of Custody:** Eliminates undocumented ditch cleaning where scooped debris is abandoned on embankments and washed back into canals during the next downpour.
- **Reconciled Mass Balance (Zero Greenwashing):** Mandates calibrated digital scale weighing (kg) segregated into PET, flexible films, silt/sediment, and organic fractions before disposal or MRF transfer.
- **Sub-Second Low-Bandwidth Resilience:** Employs client-side canvas compression downscaling 8MB mobile camera uploads to ~120KB in <100ms, enabling real-time operation over congested 3G/4G cellular networks.
- **Integrated Stormwater Banking (Rain Bank):** Couples drainage clearing with rooftop rainwater catchment modeling to calculate volumetric peak-flow attenuation for campus and municipal facilities.

2. Local Hydrological Context & Drainage Bottlenecks in CDO

Cagayan de Oro City, situated in Northern Mindanao along Macajalar Bay, is topographically characterized by steep mountainous headwaters draining through high-density coastal floodplains. The city has experienced devastating flood events, most notably Tropical Storm Sendong (Washi) in 2011, and continues to suffer severe recurring localized inundations during routine afternoon convective downpours and seasonal southwest monsoons (Habagat).

Critical Urban Flood Hotspots and Chokepoints:

1. **Bitan-ag Creek Drainage Corridor:** The critical urban drainage spine running through Camaman-an, Cogon Market, Claro M. Recto Avenue (Limketkai / USTP basin), Lapasan, and discharging into Macajalar Bay. It receives runoff from over 1,200 hectares of dense urban watershed.
2. **CM Recto / USTP Catchment Depression:** A notorious low-lying basin where arterial roads submerge under waist-deep water within 20 to 30 minutes of heavy precipitation, halting inter-city transit and endangering university students.
3. **Carmen, Bulua, and Kauswagan Outfalls:** Major commercial and residential sub-basins whose subterranean box culverts and curb inlets are routinely paralyzed by unsegregated solid waste accumulation.

Operational Dimension	Conventional Municipal / Campus Practice	TubigLoop Closed-Loop Architecture
Incident Reporting	Scattered citizen posts on social media or manual logbooks without GPS or structural metadata.	Mobile PWA with automated Vision AI material tagging, GPS auto-center, and forced pin verification.
Triage & Prioritization	Ad-hoc, subjective dispatching based on political noise or crisis complaints.	Transparent mathematical index (0–5) combining obstruction severity, disruption impact, and wait time.
Field Evidence	No photo verification; work orders marked 'cleared' with zero objective visual proof.	Mandatory Before & After photo capture with side-by-side reviewer inspection and continuous zoom audit.
Material Accounting	Debris treated as unmeasured bulk 'garbage' dumped directly into landfills or left roadside.	Calibrated scale weighing by fraction (kg of PET, film, silt, organic) with strict mass-balance accounting.
Circular Handover	Zero linkage to MRFs; recyclables contaminate waterways or municipal open disposal sites.	Formal MRF custody transfer receipts; auditable diverted plastic mass telemetry with 1-click JSON export.

3. Governance & Five-Tier Role-Based Access Control (RBAC)

TubigLoop enforces ironclad security boundaries and user segregation backed by PostgreSQL Row Level Security (RLS) and Next.js server-side route guards. Each stakeholder operates within a dedicated operational interface tailored to their institutional responsibilities:

Role	Primary Stakeholders	Allowed Route Scopes	Core System Responsibilities
Reporter	Citizens, university students, barangay tanods, campus occupants	/report /map	Submits photographic obstruction evidence, triggers Gemini AI material scan, verifies street pin via map, receives tracking receipt.
Reviewer	Campus maintenance supervisors, LGU CLENRO / CDRRMD dispatchers	/review /map /dashboard	Inspects unverified queue, executes 1.0x–3.5x image zoom inspection, assigns 0–5 priority score, flags duplicates, dispatches work orders.
Collector	Authorized sanitation crew, physical plant maintenance, sweepers	/tasks /tasks/[id] /map	Navigates to site, performs physical clearing, logs mandatory Before/After photo evidence, inputs calibrated scale weights (kg) by material fraction.
Recycler	Accredited MRF operators, junk shop cooperatives, upcyclers	/dashboard /tasks /map	Inspects diverted batches of PET and clean plastics, signs digital custody transfer receipts, accepts net verified kilograms for circular processing.
Admin	University Director of Physical Plant, City Drainage Chief, System Engineer	All routes (inc. /admin, /rain-bank)	User role governance (promote/demote), campus site configuration, raw audit log inspection, CSV/JSON export, system telemetry monitoring.

4. The End-to-End Operational Lifecycle

The TubigLoop lifecycle bridges community-driven detection with municipal accountability through a seven-step closed loop. Every physical event is mirrored by a cryptographically secured database transaction:

1. Reporter	2. Map GIS	3. Reviewer	4. Collector	5. Recycler	6. Admin
Photo + AI Tag	Hotspot Aggregation	Triage & Score	Weigh & Clear	MRF Handover	Mass Balance

Step 1: Rapid Multimodal Photo Reporting (/report)

A citizen or student encounters an obstructed canal or culvert. They snap a photo via direct camera or gallery. The browser instantly downscales the image via HTML5 Canvas (<100ms) and initiates an asynchronous Google Gemini Flash Lite vision cascade. Within ~1.5s, the AI categorizes dominant waste fractions (e.g., PET bottles, silt, organic waste). The user retains total manual control to add/remove custom material tags and classify the drainage type (Curb Inlet, Catch Basin, Open Canal, Box Culvert, Creek Outfall). The user pins the exact location on an interactive Leaflet map; OpenStreetMap Nominatim reverse geocodes coordinates to a human-readable street and barangay. Upon submission, a full-screen celebration confirms server synchronization and generates an auditable tracking receipt.

Step 2: Interactive GIS Hotspot Mapping (/map)

Submitted reports immediately register on the central Leaflet GIS map. Markers are color-coded by operational status (Pending Verification = Amber, Dispatched/In Progress = Blue, Cleared = Emerald). GIS clustering aggregates dense urban areas (such as Cogon Market or Lapasan Highway), enabling dispatchers and city planners to spot macro-level drainage failures at a glance.

Step 3: Review, Image Inspection & Priority Scoring (/review)

Maintenance dispatchers inspect incoming queues. To prevent misjudgments caused by tiny thumbnails, reviewers can click an inspection eye icon to open an evidence modal featuring 1.0x to 3.5x continuous image zoom and complete reporter notes. The reviewer confirms validity, filters out duplicates, and scores the incident using a transparent 0–5 priority index.

Step 4: Maintenance Work Order Execution (/tasks)

The verified report converts into a field work order assigned to a specific cleaning crew. The crew accesses their mobile task dashboard, navigates to the GPS pin, and executes the physical clearing using appropriate tools (silt scoops, trash rakes).

Step 5: Mandatory Dual-Photo Verification & Scale Weighing (/tasks/[id])

The collection crew is strictly barred from closing a task without verifiable proof. They must upload an 'After' clearance photo taken from the identical vantage point and enter measured scale weights (kg) across segregated bins: PET Plastic, Flexible Packaging, Silt/Sediment, and Organic/Residual Waste. Qualitative estimates are strictly barred from numerical ledger sums.

Step 6: Recycler Custody Transfer & Handover Sign-Off (/dashboard)

Recovered recyclable plastics are not dumped into landfills. They are routed to accredited local junk shops or barangay MRFs. The recycler verifies the weight on their own scale and signs off on the digital custody transfer, producing an auditable digital receipt.

Step 7: Reconciled Mass Balance & Public Telemetry (/dashboard & /rain-bank)

The platform aggregates citywide data into an unassailable mass-balance ledger. The dashboard proves exactly how many kilograms were diverted from Macajalar Bay into circular manufacturing versus residual landfill disposal. Concurrently, the Rain Bank module models rooftop catchment volumes to relieve peak stormwater pressure during cloudbursts.

5. Technical Implementation & Software Engineering

TubigLoop is architected as a resilient, production-ready enterprise Progressive Web App (PWA) built on modern Next.js 16 serverless infrastructure, Supabase PostgreSQL, and Google Gemini Multimodal AI. The system follows strict engineering tenets: zero client-side optimistic fabrication, server-verified truth, and zero UI gradients.

Architecture Layer	Technology	Version / Specification	Operational Implementation in TubigLoop
Application Framework	Next.js App Router	Next.js 16.4.2 / Turbopack	Server Components, Server Actions for zero-leakage API calls, Route Handlers, automatic code-splitting.
UI Runtime & Lang	React 19 / TypeScript 5	React 19.3.0 / Strict TS	Modern concurrent rendering, typed database schema interfaces, zero runtime type errors.
Styling System	Tailwind CSS v4	Tailwind v4 (Zero Gradients)	Official 4-color palette tokens (#30AFFF, #92EEFF, #D8FFC5, #C4F7CA), crisp border contrast, flat cards.
Database & Security	Supabase PostgreSQL	Postgres 15+ / SSR Cookies	Row Level Security (RLS) policies, session-based auth via @supabase/ssr, immutable audit log tables.
Multimodal Vision AI	Google Gemini API	gemini-3.5-flash-lite / flash-lite-latest	Dual-model cascade with 6s timeout, structured JSON schema response, automatic human fallback.
GIS & Geocoding	Leaflet & OSM Nominatim	Leaflet 1.9.4 / OSM Tiles	Zero-cost open mapping, GPS coordinate capture, reverse geocoding to CDO streets and barangays.
Client Processing	HTML5 Canvas API	Native Browser Canvas	Pre-upload image downscaling to 960x960 @ 0.72 quality in <100ms; reduces upload payload by ~98%.

Low-Bandwidth Mobile Optimization Engine:

In field disaster conditions or congested cellular environments (common along flooded corridors in CDO), standard 8MB to 12MB smartphone photographs fail to upload, causing critical report abandonment. TubigLoop implements an in-memory client-side HTML5 Canvas transformation pipeline. The raw bitmap is scaled to a bounding box of 960x960 pixels at 0.72 JPEG compression within 80 milliseconds on the user's device. The resulting ~120KB payload transmits reliably even over 2G/3G links and substantially reduces serverless inference latency on Google Gemini.

Gemini Multimodal Fast-Cascade Architecture:

The backend AI engine (src/lib/gemini.ts) executes a hardened multi-tier strategy: it initiates a primary inference request to gemini-3.5-flash-lite with a strict 6.0-second timeout. If network degradation occurs or rate limits are met, it seamlessly cascades to gemini-flash-lite-latest. If all cloud AI endpoints are unreachable, the application degrades gracefully without breaking the user experience, falling back to manual chip selection.

6. Mathematical Formulations & Anti-Greenwashing Metrics

To satisfy rigorous academic, municipal, and international donor audit requirements, TubigLoop repudiates subjective claims. All triage rankings, recycling recovery rates, and rainwater offsets are derived from deterministic equations:

A. Transparent Priority Triage Scoring Algorithm (Range: 0 to 5)

Verified drainage reports are assigned an integer priority score *P* evaluated as the sum of three independent variables:
• **Obstruction Score (S_obs in {0, 1, 2}):** 0 = Minimal debris; 1 = Partial constriction (<50%); 2 = Total hydraulic blockage (>50%).
• **Disruption Score (S_dis in {0, 1, 2}):** 0 = Low-impact ditch; 1 = Affects pedestrian walkway; 2 = Critical arterial road or campus entrance.
• **Time Decay Penalty (S_time in {0, 1}):** +1 if report has remained unassigned for >48 hours post-verification; 0 otherwise.

Priority Index Equation: P = S_obs + S_dis + S_time
• **P in [0, 1]: Low Priority** (Scheduled routine maintenance within 7 days)
• **P in [2, 3]: Medium Priority** (Active dispatch within 48 hours)
• **P in [4, 5]: High Priority** (Emergency rapid response within 6–12 hours)

B. Closed-Loop Mass Balance Reconciliation (Kilogram Accounting)

Greenwashing is prevented by enforcing strict mathematical equivalence between field collection scales and downstream disposition:

1. Total Wet Mass Collected:
M_total = M_PET + M_film_plastic + M_silt + M_organic + M_residual

2. Verified Circular Diversion Rate (Div_Rate %):
Div_Rate % = (M_accepted_MRF / M_total) * 100%

Rule: M_accepted_MRF is strictly populated ONLY when an authorized Recycler counter-signs a digital handover receipt. Unweighed estimations or unconfirmed batches remain classified as 'Awaiting Disposition' and are excluded from diversion claims.

C. Rain Bank Rooftop Stormwater Retention Model

The Rain Bank module models peak-flow runoff reduction by calculating harvestable volume from campus rooftops:

Harvested Stormwater Volume (Liters):
V_harvested = A_roof (sq.m) * R_event (mm) * C_runoff * Filter_Efficiency
Where: A_roof = Catchment footprint in sq. meters; R_event = Localized precipitation depth (e.g. 50mm downpour); C_runoff = Surface runoff coefficient (0.90 for corrugated GI sheet / 0.85 for concrete tile); Filter_Efficiency = First-flush filter efficiency factor (typically 0.90).

7. Pilot Implementation Framework: USTP & Bitan-ag Corridor

The initial deployment is tailored for the **CEIW 2026** Green Schools & Learning challenge, establishing a live demonstration pilot anchored at the University of Science and Technology of Southern Philippines (USTP) Cagayan de Oro campus, coupled with the municipal drainage artery along Claro M. Recto Avenue.

Stakeholder Entity	Operational Role	Institutional Contribution & Value Realized
USTP CDO Administration & Physical Plant Office (PPO)	Campus Host & Maintenance Execution	Deploys ground maintenance sweepers as Collectors; uses TubigLoop to monitor 18 campus drainage catchments and curb inlets; protects USTP grounds from recurrent flood shutdown.
USTP Student Council & Engineering Orgs	Civic Reporters & Campus Monitors	Organizes student reporting brigades; logs geotagged photos during heavy downpours; provides active user feedback on mobile UI.
City Local Environment & Natural Resources Office (CLENRO)	LGU Solid Waste Authority	Aligns collection telemetry with City Ecological Solid Waste Management programs; audits diversion rates from Bitan-ag Creek.
City Disaster Risk Reduction & Management Dept (CDRRMD)	Flood Emergency Coordination	Monitors real-time GIS map of severe drainage chokepoints along CM Recto Avenue during monsoon alerts to deploy flood extraction pumps.
Accredited Junk Shops & MRF Cooperatives (Lapasan/Puntod)	Circular Recovery Offtakers	Acts as certified Recyclers on the platform; receives sorted PET bottles and HDPE containers; signs digital transfer receipts for circular monetization.

Six-Month Phased Rollout Schedule:

Phase	Timeline	Milestones & Deliverables
Phase I: Setup & Baseline	Month 1	Deploy Next.js 16 PWA on Vercel; configure Supabase database RLS schemas; calibrate Gemini Flash Lite cascade; map USTP campus drainage outfall coordinates.
Phase II: Campus Alpha	Month 2	Onboard 50 USTP student-leaders and 10 PPO maintenance staff; conduct controlled trials of photo reporting, scale weighing, and triage dispatching.
Phase III: MRF Integration	Month 3	Onboard 2 partner junk shops in Lapasan; establish digital custody transfer sign-offs; calibrate digital hanging crane scales for collection crews.
Phase IV: Monsoon Beta	Months 4–5	Execute live field trials during high-precipitation monsoon events; monitor CM Recto artery; validate mass-balance telemetry under heavy load (>500 reports/week).
Phase V: Audit & Policy	Month 6	Publish comprehensive mass-balance audit; deliver pilot policy memorandum to CDO City Council and DOST; package open API specifications.

8. Philippine Policy & Statutory Alignment

A compelling concept paper must substantiate compliance with prevailing Philippine environmental and disaster management mandates. TubigLoop directly addresses the operational mandates of four major statutory frameworks:

1. **Republic Act 9003 (Ecological Solid Waste Management Act of 2000):**
   Mandates segregation at source and minimum 25% waste diversion from disposal facilities. TubigLoop provides LGUs and universities with verifiable, kilogram-calibrated digital records of diverted materials, directly supporting barangay MRF audits.
2. **Republic Act 11898 (Extended Producer Responsibility Act of 2022 - EPR Law):**
   Obliges large enterprise product manufacturers to account for plastic packaging recovery. TubigLoop's immutable custody transfer receipts and PET/film segregation records serve as uncompromised third-party verification for corporate EPR compliance credits.
3. **Republic Act 10121 (Philippine DRRM Act of 2010):**
   Prioritizes proactive disaster risk reduction over reactive disaster response. By identifying and clearing drainage chokepoints *before* monsoon downpours peak, TubigLoop operationalizes institutional pre-disaster flood mitigation.
4. **CDO City Ordinance No. 13378-2018 (Integrated Solid Waste Management Ordinance):**
   Prohibits the dumping of refuse into waterways and open drainage canals. The platform equips city inspectors and community brigades with geotagged photographic evidence to enforce municipal sanitation standards.

9. Anticipated Impacts & Target Key Performance Indicators

Concept papers submitted to academic bodies, DOST, or international granting entities require rigorous, quantifiable KPIs. The following performance matrix defines the benchmark evaluation criteria for the TubigLoop pilot deployment:

Impact Dimension	Target Metric / Indicator	Baseline (Conventional)	Projected Pilot Target
Disaster Resilience	Drainage obstruction clearance SLA (Report to Verified Cleared)	7 to 21 days (or unaddressed)	**<= 24 hours** for High Priority (P>=4); **<= 72 hours** for Medium Priority.
Flood Hazard Reduction	Drainage hydraulic cross-section availability along Bitan-ag	40%–60% choked by silt and plastic	**>= 85% clear cross-section** maintained across targeted pilot corridor.
Circular Recovery	Diverted plastic mass recovered from drainage corridors	0% (100% dumped or washed to sea)	**>= 3,500 kg** of PET and rigid plastics diverted to MRFs over 6-month pilot.
Mass-Balance Integrity	Audit discrepancy between collection and disposition	Unmeasured / complete black hole	**< 2.0% discrepancy** between scale weights and custody transfers.
Community Mobilization	Active civic/student reporters and validated submissions	Passive / reactive Facebook rants	**>= 250 active reporters**; **>= 1,000 verified photographic reports** logged.
Stormwater Buffering	Rain Bank rooftop runoff modeled and retained	0 liters retained (100% flash runoff)	**>= 150,000 liters** of localized rainwater retention capacity modeled.

10. Directory of Acronyms & Technical Glossary

Acronym / Term	Definition & Context in TubigLoop
CDO	Cagayan de Oro City, Northern Mindanao, Philippines (Target deployment municipality).
CEIW 2026	Circular Economy & Green Innovation Pitching Competition / Challenge program.
CLENRO	City Local Environment and Natural Resources Office (CDO's municipal solid waste authority).
CDRRMD	City Disaster Risk Reduction and Management Department of Cagayan de Oro.
USTP	University of Science and Technology of Southern Philippines (CDO Campus).
MRF	Materials Recovery Facility (Facility for segregating, baling, and processing recyclables).
PWA	Progressive Web App (App that works seamlessly across iOS/Android browsers without app store gatekeeping).
RLS	Row Level Security (PostgreSQL security mechanism enforcing row-by-row actor permissions).
EPR	Extended Producer Responsibility (RA 11898 obliging brand owners to recover plastic packaging).
DRRM	Disaster Risk Reduction and Management (RA 10121 framework for climate-resilient cities).
GIS	Geographic Information System (Spatial mapping and coordinate visualization via Leaflet / OSM).
Bitan-ag Creek	The primary 7-kilometer urban drainage tributary running through the center of Cagayan de Oro.
Rain Bank	Integrated stormwater retention calculator modeling non-potable rainwater harvesting to abate flash runoff.

Summary Note for Proposal Authors:
This briefing provides the complete systemic architecture for authoring academic concept papers, grant proposals, and LGU memoranda of agreement. The software implementation is fully developed, operational, and verified in codebase repository JRSstuff/TubigLoop. All features referenced—including client-side canvas downscaling, Gemini Flash Lite vision categorization, reverse geocoding, 1.0x–3.5x image zoom triage inspection, scale weighing inputs, and mass-balance accounting—are fully functional production modules ready for field demonstration.